

ACID_TEST_HR_DEMO

INITIAL EMPLOYEE DATA

```
41 • SELECT 'INITIAL EMPLOYEE DATA' AS info;  
42 • SELECT * FROM employees;
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell
employee_id	employee_name	department	salary	leave_balance
1	Aung Aung	IT	800000.00	10
2	Su Su	HR	600000.00	8
3	Kyaw Kyaw	Finance	900000.00	12
4	Hla Hla	Admin	550000.00	6
5	Mya Mya	IT	750000.00	9
6	Ko Ko	Operations	700000.00	11
* NULL	NULL	NULL	NULL	NULL

ATOMICITY TEST - BEFORE SUCCESSFUL TRANSACTION

```
55 • SELECT 'ATOMICITY TEST - BEFORE SUCCESSFUL TRANSACTION' AS info;  
56 • SELECT * FROM employees WHERE employee_id = 1;  
57
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Ce
employee_id	employee_name	department	salary	leave_balance
1	Aung Aung	IT	800000.00	10
* NULL	NULL	NULL	NULL	NULL

SUCCESSFUL ATOMIC TRANSACTION

START TRANSACTION;

UPDATE employees

SET salary = salary + 50000

WHERE employee_id = 1;

UPDATE employees

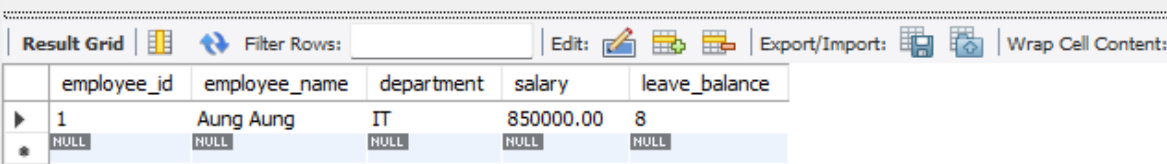
SET leave_balance = leave_balance - 2

WHERE employee_id = 1;

COMMIT;

ATOMICITY TEST - AFTER SUCCESSFUL COMMIT

```
74 • SELECT 'ATOMICITY TEST - AFTER SUCCESSFUL COMMIT' AS info;
75 • SELECT * FROM employees WHERE employee_id = 1;
```



	employee_id	employee_name	department	salary	leave_balance
▶	1	Aung Aung	IT	850000.00	8
*	NULL	NULL	NULL	NULL	NULL

ATOMICITY TEST - BEFORE FAILED TRANSACTION

```

82 • SELECT 'ATOMICITY TEST - BEFORE FAILED TRANSACTION' AS info;
83 • SELECT * FROM employees WHERE employee_id = 2;

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell C
employee_id	employee_name	department	salary	leave_balance
2	Su Su	HR	600000.00	8
NULL	NULL	NULL	NULL	NULL

START TRANSACTION;

UPDATE employees

SET salary = salary + 50000

WHERE employee_id = 2;

-- Mistake / unexpected issue

-- This update affects no valid intended record

UPDATE employees

SET leave_balance = leave_balance - 100

WHERE employee_id = 999;

ROLLBACK;

ATOMICITY TEST - AFTER ROLLBACK

```

99 • SELECT 'ATOMICITY TEST - AFTER ROLLBACK' AS info;
100 • SELECT * FROM employees WHERE employee_id = 2;

```

Result Grid	Filter Rows:	Edit:	Export/Import:	V
employee_id	employee_name	department	salary	leave_balance
2	Su Su	HR	600000.00	8
NULL	NULL	NULL	NULL	NULL

CONSISTENCY

CONSISTENCY TEST - BEFORE VALID UPDATE

```

115 • SELECT 'CONSISTENCY TEST - BEFORE VALID UPDATE' AS info;
116 • SELECT * FROM employees WHERE employee_id = 3;
117

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap C
employee_id	employee_name	department	salary	leave_balance
3	Kyaw Kyaw	Finance	900000.00	12
NULL	NULL	NULL	NULL	NULL

START TRANSACTION;

UPDATE employees

SET leave_balance = leave_balance - 3

WHERE employee_id = 3

AND leave_balance >= 3;

COMMIT;

CONSISTENCY TEST - AFTER VALID UPDATE

```

131 • SELECT 'CONSISTENCY TEST - AFTER VALID UPDATE' AS info;
132 • SELECT * FROM employees WHERE employee_id = 3;
133

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap C
employee_id	employee_name	department	salary	leave_balance
3	Kyaw Kyaw	Finance	900000.00	9
NULL	NULL	NULL	NULL	NULL

CONSISTENCY TEST - BEFORE INVALID ATTEMPT

```

138 • SELECT 'CONSISTENCY TEST - BEFORE INVALID ATTEMPT' AS info;
139 • SELECT * FROM employees WHERE employee_id = 4;

```

Result Grid					
Filter Rows:					
Edit:					
Export/Import:					
Wrap Cell Content:					
	employee_id	employee_name	department	salary	leave_balance
▶	4	Hla Hla	Admin	550000.00	6
*	NULL	NULL	NULL	NULL	NULL

START TRANSACTION;

UPDATE employees

SET leave_balance = leave_balance - 20

WHERE employee_id = 4

AND leave_balance >= 20;

COMMIT;

CONSISTENCY TEST - AFTER INVALID ATTEMPT (NO CHANGE EXPECTED)

```

150 • SELECT 'CONSISTENCY TEST - AFTER INVALID ATTEMPT (NO CHANGE EXPECTED)' AS info;
151 • SELECT * FROM employees WHERE employee_id = 4;

```

Result Grid					
Filter Rows:					
Edit:					
Export/Import:					
Wrap Cell Content:					
	employee_id	employee_name	department	salary	leave_balance
▶	4	Hla Hla	Admin	550000.00	6
*	NULL	NULL	NULL	NULL	NULL

DURABILITY

DURABILITY TEST - BEFORE COMMIT

```

4 • SELECT 'DURABILITY TEST - BEFORE COMMIT' AS info;
5 • SELECT * FROM employees WHERE employee_id = 2;

```

Result Grid

Filter Rows:

Edit:

Export/Import:

	employee_id	employee_name	department	salary	leave_balance
▶	2	Su Su	HR	600000.00	8
*	NULL	NULL	NULL	NULL	NULL

START TRANSACTION;

UPDATE employees

SET salary = 950000

WHERE employee_id = 2;

COMMIT;

DURABILITY TEST - AFTER COMMIT

```

15 • SELECT 'DURABILITY TEST - AFTER COMMIT' AS info;
16 • SELECT * FROM employees WHERE employee_id = 2;
17

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap
employee_id	employee_name	department	salary	leave_balance
2	Su Su	HR	950000.00	8
NULL	NULL	NULL	NULL	NULL

PROMOTION TEST - BEFORE SUCCESSFUL PROMOTION

```

38 • SELECT 'PROMOTION TEST - BEFORE SUCCESSFUL PROMOTION' AS info;
39 • SELECT * FROM employees WHERE employee_id = 1;
40

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell C
employee_id	employee_name	department	salary	leave_balance
1	Aung Aung	IT	850000.00	8
NULL	NULL	NULL	NULL	NULL

START TRANSACTION;

UPDATE employees

SET salary = salary + 150000

WHERE employee_id = 1;

UPDATE employees

SET department = 'Senior IT'

WHERE employee_id = 1;

UPDATE employees

SET leave_balance = leave_balance + 2

WHERE employee_id = 1;

COMMIT;

PROMOTION TEST - AFTER SUCCESSFUL PROMOTION

```

60 • SELECT 'PROMOTION TEST - AFTER SUCCESSFUL PROMOTION' AS info;
61 • SELECT * FROM employees WHERE employee_id = 1;
62

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wr
employee_id	employee_name	department	salary	leave_balance
1	Aung Aung	IT	850000.00	8
NULL	NULL	NULL	NULL	NULL

PROMOTION TEST - BEFORE FAILED PROMOTION

```
67 • SELECT 'PROMOTION TEST - BEFORE FAILED PROMOTION' AS info;  
68 • SELECT * FROM employees WHERE employee_id = 6;
```

Result Grid					
Filter Rows:					
Edit:					
Export/Import:					
Wrap Cell					
	employee_id	employee_name	department	salary	leave_balance
▶	6	Ko Ko	Operations	700000.00	11
*	NULL	NULL	NULL	NULL	NULL

START TRANSACTION;

UPDATE employees

SET salary = salary + 150000

WHERE employee_id = 6;

UPDATE employees

SET department = 'Senior Operations'

WHERE employee_id = 6;

-- Suppose this is wrong

UPDATE employees

SET leave_balance = leave_balance + 2

WHERE employee_id = 999;

ROLLBACK;

PROMOTION TEST - AFTER FAILED PROMOTION ROLLBACK


```
87 • SELECT 'PROMOTION TEST - AFTER FAILED PROMOTION ROLLBACK' AS info;
```

```
88 • SELECT * FROM employees WHERE employee_id = 6;
```

SQL

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell C
employee_id	employee_name	department	salary	leave_balance
6	Ko Ko	Operations	700000.00	11
* NULL	NULL	NULL	NULL	NULL

FINAL TEST DATA VIEW

FINAL EMPLOYEE DATA

```
93 • SELECT 'FINAL EMPLOYEE DATA' AS info;
```

```
94 • SELECT * FROM employees;
```

SQL

Result Grid

Filter Rows:

Edit:

Export/Import:

	employee_id	employee_name	department	salary	leave_balance
▶	1	Aung Aung	IT	850000.00	8
	2	Su Su	HR	950000.00	8
	3	Kyaw Kyaw	Finance	900000.00	9
	4	Hla Hla	Admin	550000.00	6
	5	Mya Mya	IT	750000.00	9
	6	Ko Ko	Operations	700000.00	11
*	NULL	NULL	NULL	NULL	NULL